Technical Report — Shock Squad

1. Problem Understanding

Our objective is to flag fraudulent transactions while disrupting as few legitimate customers as possible. Only 353 of 20,000 training transactions are fraud (1.8%, about 1 in 57), so a model that never flags anything is 98.2% accurate yet catches no fraud. The competition scores **PR-AUC** on predicted fraud probabilities, so we optimise how well the model **ranks** fraud above legitimate transactions; a random guess scores only 0.018.

2. Data Processing & Feature Engineering

There are no duplicates, and 975 rows (4.9%) have one or two missing values with no clear fraud signal, so we filled numbers with the **median** and categories with the **most common value**, learned inside each training fold. We removed `id` as it is an identifier. Exploration found strong single signals: **new devices** (7.5% fraud vs 1.2%), luxury and cash-transfer merchants, late night (23:00–03:00), large amounts and young accounts. We engineered **log amount**, **amount relative to 24-hour spend**, **log account age**, a **cyclical hour** and a **night flag**, plus five features for the final model: **average spend per transaction**, **amount relative to that average**, **share of 24-hour activity in the last hour** and two **new-device interactions**. Our `make_features` function turns bad values into missing ones, clips values to the training range and needs no `id` column.

3. Model Selection & Justification

Every model was scored on the same 15 folds. **Logistic regression** (PR-AUC 0.228) beat a "flag every new device" rule (0.040), class-weighted logistic regression (0.215), **Gradient Boosting** (0.203), Gradient Boosting with **SMOTE** oversampling (0.198) and **Random Forest** (0.178). Extra Trees, LightGBM, splines, interaction terms and linear–tree blends did not beat it reliably either, and leaving out our engineered features dropped PR-AUC to 0.207. The final model is **logistic regression with stronger regularisation (`C=0.1`) and the five extra features: PR-AUC 0.236**, better than the baseline in 12 of 15 folds. We kept the simplest and most explainable model; its strongest risk factors are cash transfers, luxury merchants and new devices.

4. Evaluation Strategy

We used **5-fold stratified cross-validation repeated 3 times** (15 folds of about 70 frauds each) and report mean ± standard deviation. We submit **fraud probabilities**; `predict()` also flags at a threshold of 0.11 chosen for F1, and choosing it strictly inside each fold gave the same F1. Four checks show the model can be trusted. **Learning curve:** training and validation PR-AUC meet (0.246 vs 0.235), so it neither overfits nor underfits, while Extra Trees overfits (0.443 vs 0.227). **Calibration:** scores match real fraud rates (top tenth: 8.9% predicted, 8.6% observed). **Usefulness:** the top 1% of scores holds 23% of all fraud and the top 5% holds 39%. **Reproducibility:** one notebook runs in training mode (trains and saves `model.pkl`) or prediction mode (no training data, as on the platform), tested with the platform's exact library versions on shuffled rows without `id`.

5. Limitations & Risks

Bias: fraud in Singapore is caught only 9% of the time, against 26–71% in other countries, and false-alarm rates differ about 6× across countries; card-present and mobile-app fraud is also caught far less often. Removing `country` barely helps (5.7× to 5.1×), so the gap comes from how transactions differ, not the label; separate thresholds per group would equalise it, but need a business decision on which fairness matters. **Data ceiling:** the validation curve is flat, so better detection needs new information such as account history, not a bigger model. **Shift:** the model flags 2.8% of the public test set against 1.7% of the training data, as it has more new devices and larger amounts; we cannot verify this without labels.